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India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Session : Power Markets & Carbon Markets

Topic: Impact of Energy Transition on Power Markets

Presented By

Rohit Bajaj, JMD, Indian Energy Exchange

IEX: India's premier technology-led energy marketplace

- Nation wide, automated and transparent trading platform for physical delivery of electricity, renewables & certificates
- Commenced operations in 2008; CERC regulated
- Publicly listed company – 2017 (NSE and BSE)
- ISO 9001:2015, ISO 27001:2013, ISO 14001:2015
- IEX is on the Board of Association of Power Exchanges (APEx) where 40 Global exchanges are members
- IEX has diversified into Gas Exchange (IGX)
- IEX is member of 13 State Regulatory Commission advisory committees
- Regularly organize Capacity building workshop for state utilities and other participants

Robust Ecosystem

7900+ Registered participants	**4500+** Commercial & Industries
1500+ RE Generators & Obligated Entities	**600+** Generators
55+ Discoms (all)	**100+** ESCert Entities

Leading the future of energy with innovation and technology

Impact of Energy Transition on Market

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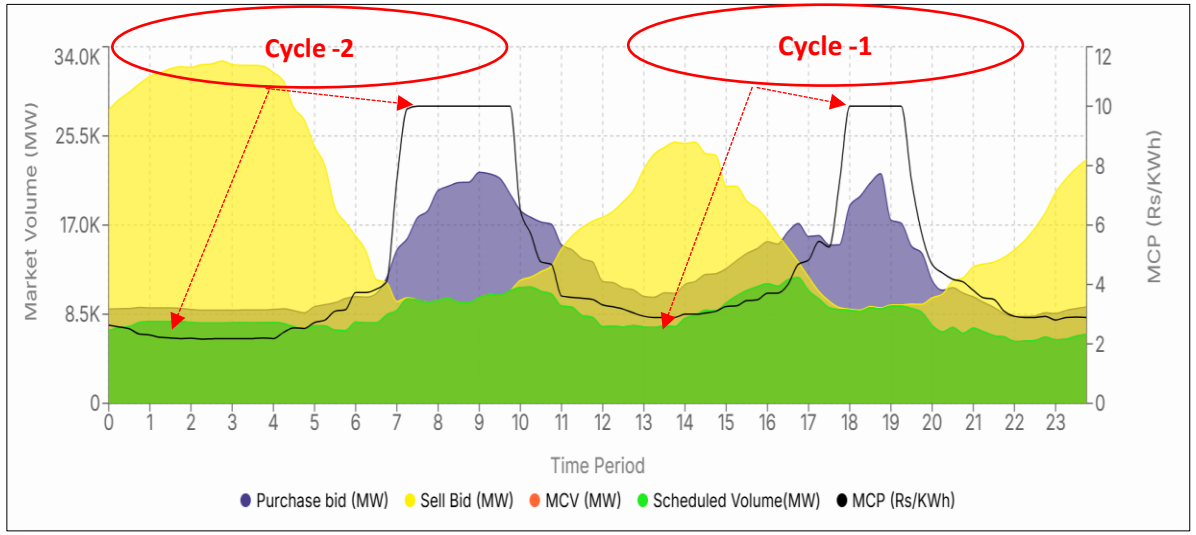
Key Trends

1 Excess Supply During Solar hrs

- ✓ Leads to extremely low MCP
- ✓ RE Curtailment
- ✓ Missing money problem for Generators serving baseload



Interventions Needed

- ✓ ***Focus on adding BESS & PSH***
  - *As Per CEA targeted addition of BESS (FY32): 47 GW/236 GWh*
  - *As per CEA, targeted addition of PSP (FY32): 27 GW/175 GWh*
  - *VGF to support addition 22 GW of BESS over next 2-3 years*
  - *Accelerate addition of Storage through Power System modelling (i.e. by citing of BESS at best location in Transmission and distribution system)*
- ✓ ***BESS Arbitrage opportunity***
  - *Strong case for market-based BESS (2-hr & 4-hr)*

Net Market Based Arbitrage opportunity

FY	2 Hrs		4 Hrs.	
	Annual Arbitrage (Rs/kWh)	No. of cycles	Annual Arbitrage (Rs/kWh)	No. of cycles
FY 24	3.80	568	3.18	491
FY 25	5.03	523	3.94	485

Net of RTE -85%; ISTS Loss, PX Fee

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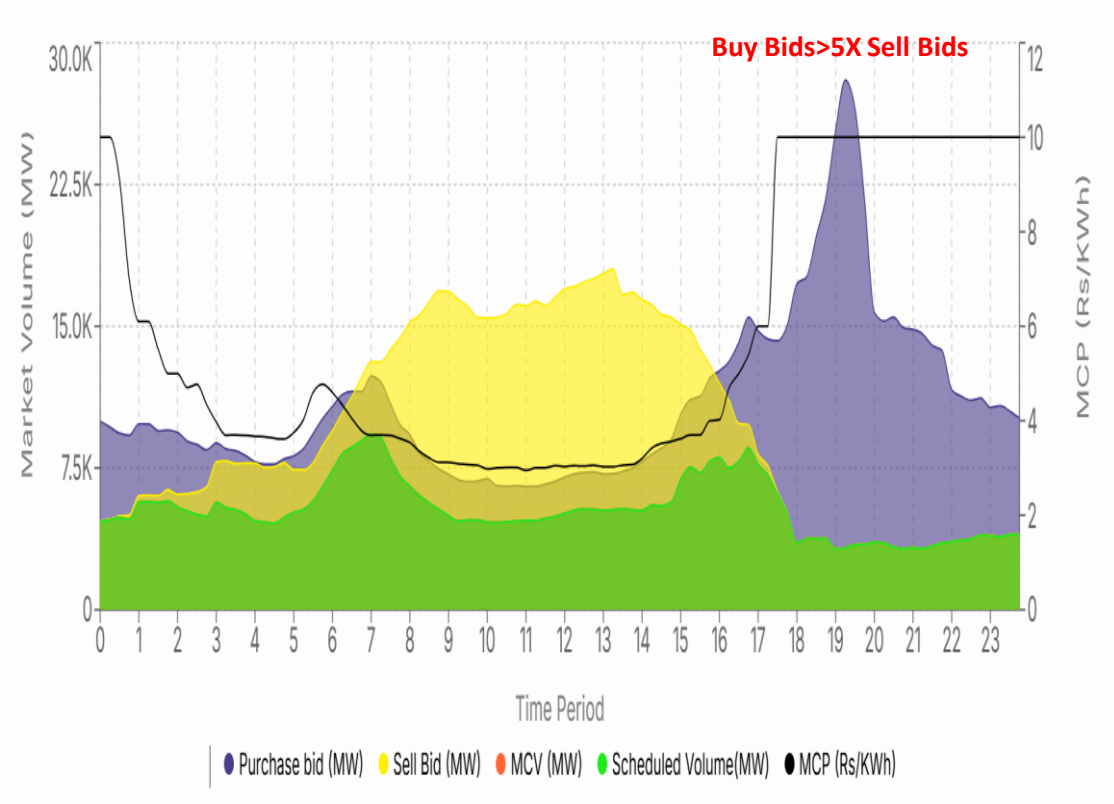
Key Trends

Intervention Needed

2 Deficit supply during Peak Hours

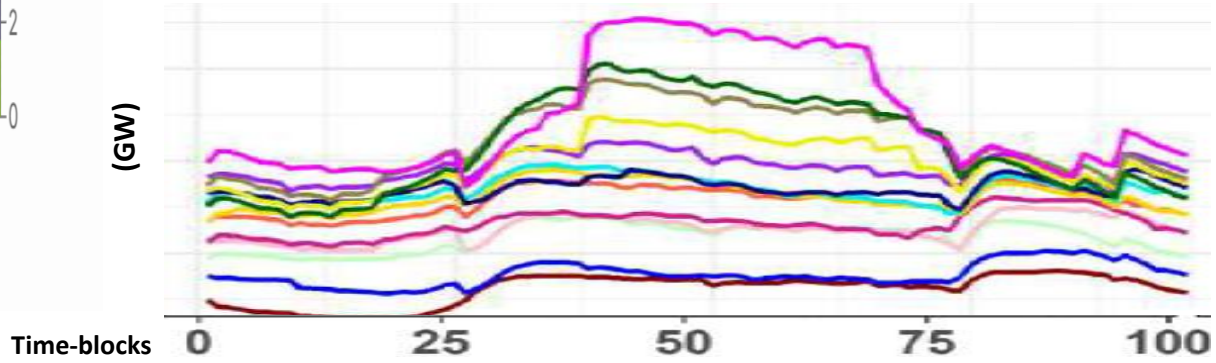


✓ ***Shifting Agri-culture load to Solar hours***

- *Total grid connected load in India is ~ 158 GW & Agri-consumption has grown at a CAGR of 5.2% over the last 10 years; (FY 24 - 255 BU)*
- *During FY19 to FY35 ~ 60 GW of Agri-load is expected to shift to Solar hours; ~ 30 GW has already been shifted & remaining should be expedited*

✓ ***Time of the day Tariff at retail level***

- *Retail tariff with broader gap b/w Solar & Non-Solar hours to incentivize consumption during solar Hrs.*
- *Linking ToD tariff with market MCP*



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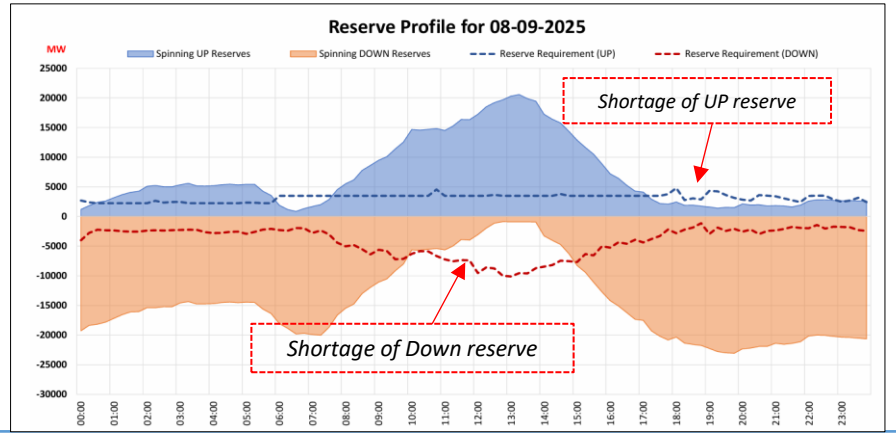
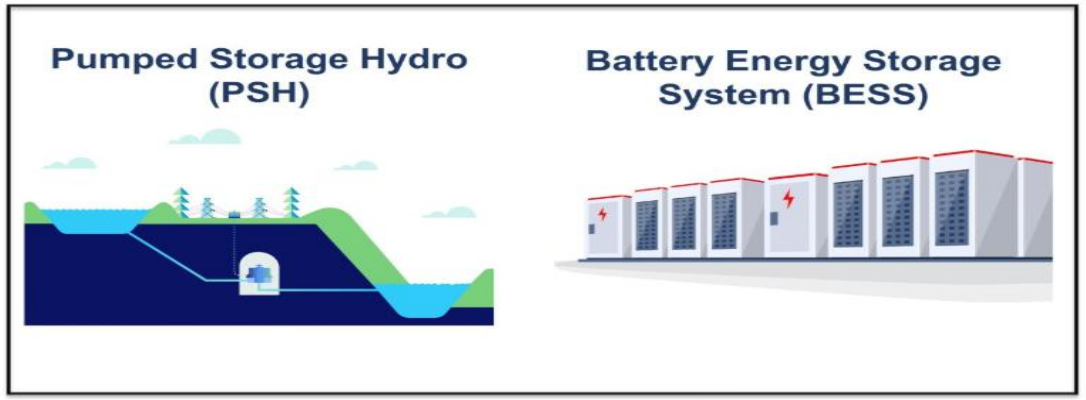
Key Trends | **Intervention needed**

3 Shortage of Ancillary Services (balancing support)

- ✓ *Limited addition of peaking and flexible resources has led to shortage of Ancillary reserves*
- ✓ *Shortage of Down reserves are higher than the UP reserves (leading to Solar curtailment)*
- ✓ *Due to higher realization in Energy market, Sellers prefer to participate in Energy market rather than Ancillary market*
- ✓ *Govt. Policies have mandated Bilateral PPAs to offer their un-scheduled / unsold power at Exchange (DAM & Real Time Market)*
- ✓ *On 8th October-25 Emergency VRE Curtailment was 325 MW (~1 MU)for wind and 13712 MW(~42.18 MU) for solar and the maximum curtailment occurred in NR Region*



- ✓ ***Focus on retrofitting of thermal stations***
  - *Thermal Stations are required to retrofit their Boilers to reduce Technical Minimum operation from 55% of TMCR (at present) to 40%*
  - *This shall increase Flexibilization which is crucial to provide Down reserve as reduce Solar curtailment*
- ✓ ***Attracting flexible products in ancillary market for grid stabilisation like-***
  - *BESS I PSP I Gas*



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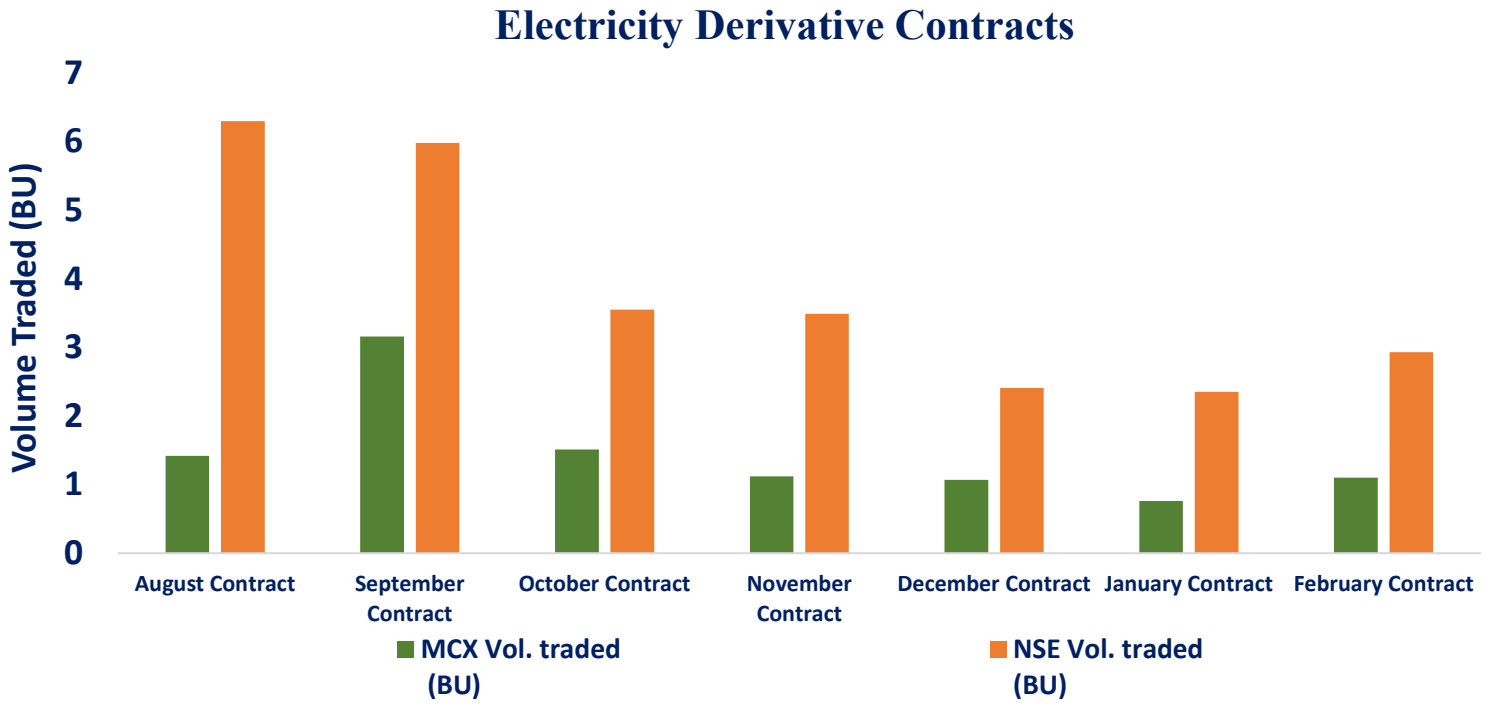
Key Trends	Intervention needed
4 Rising cost of Transmission charges ✓ Transmission cost has increased with Solar addition (currently ~ 15-20% of energy cost) ✓ To increase utilization of transmission capacity, Solar plants are mandated to install BESS to retain transmission rights for non-solar hours	✓ For Optimal Transmission capacity utilisation BESS should be integrated with Transmission system. RE-RTC capacity addition should be promoted. ✓ In Existing Solar parks – BESS could be added for Non-Solar hour utilisation. ✓ Waiver of Transmission charges to RE should be gradually withdrawn (incentivize evaluation of setting up RE near demand v/s setting up RE in rich states which requires building of ISTS)
5 Demand Trends ✓ CEA's projections shows demand to grow at the CAGR of ~ 6% over next decade ✓ Increasing co-relation with weather events ✓ Air Conditioning (climate change) ~ 40 GW (FY25) to ~ 80 GW by FY30 ✓ **Additional Demand drivers** • Data centres (10 GW of additional demand by 2030) • Data centre demand : 13 BU (FY25) to ~ 50 BU (FY30)	✓ Ensuring Resource Adequacy Planning at temporal & granular level to meet the demand profile at demand centres ✓ Pro-active planning of expansion of T&D network to avoid congestion & including BESS as integral to T&D expansion ✓ Focus on addition of Peaking resources (Thermal, PSPs, longer duration BESS) ✓ Capacity Markets for emerging technologies: Longer duration BESS, PSPs etc.

Electricity Derivative Market

A derivative is a financial contract whose value is derived from one or more underlying assets or related variables, making it a contract rather than an independent asset.



Electricity Derivatives-Due Date Rate (DDR)		
Contract	MCX-DDR(Rs./KWh)	NSE-DDR(Rs./KWh)
August	4.05	3.99
September	3.60	3.45
October	2.69	2.80
November	3.09	3.23
December	3.93	3.88
January	3.83	3.88
February	3.59	3.55

- **Different Reference Price for settlement of Electricity Futures**
  - **NSE: Volume weighted average of DAM/GDAM/HPDAM of all 3 Exchanges**
  - **MCX: Volume weighted average of DAM of IEX**

- **However, there can be various derivative products that can consider different Reference Price like SCED, RTM, Ancillary etc**

Way Forward

Immediate Focus

- ✓ **Optimizing Power Procurement through Exchanges:** DISCOMs and C&I consumers can leverage power exchanges to lower procurement costs. Advanced optimization tools and consultancy support can further enhance procurement efficiency in some cases.
- ✓ **Green Power Procurement through Exchanges:** Liquidity in the green power market is increasing, with the green segment gradually moving closer to real-time trading (Green RTM).

Future developments

- ✓ **Shorter Trading Intervals:** Transition towards 5 mins scheduling to enable more granular/ responsive market operations.
- ✓ **Reduced Gate Closure Time:** Gate closure time for RTM to decrease from 60 mins to 30 mins, improving market responsiveness.
- ✓ **Ancillary Market Redesign:** Greater emphasis on improving liquidity and shifting towards market-based procurement of secondary and tertiary ancillary services.
- ✓ **Peer to Peer Trading:** Development of decentralized markets involving aggregators, P2P trading, and flexibility markets.
- ✓ **Short-Term Capacity Market:** Mechanisms to encourage emerging technologies such as BESS, Peaking contracts.

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)